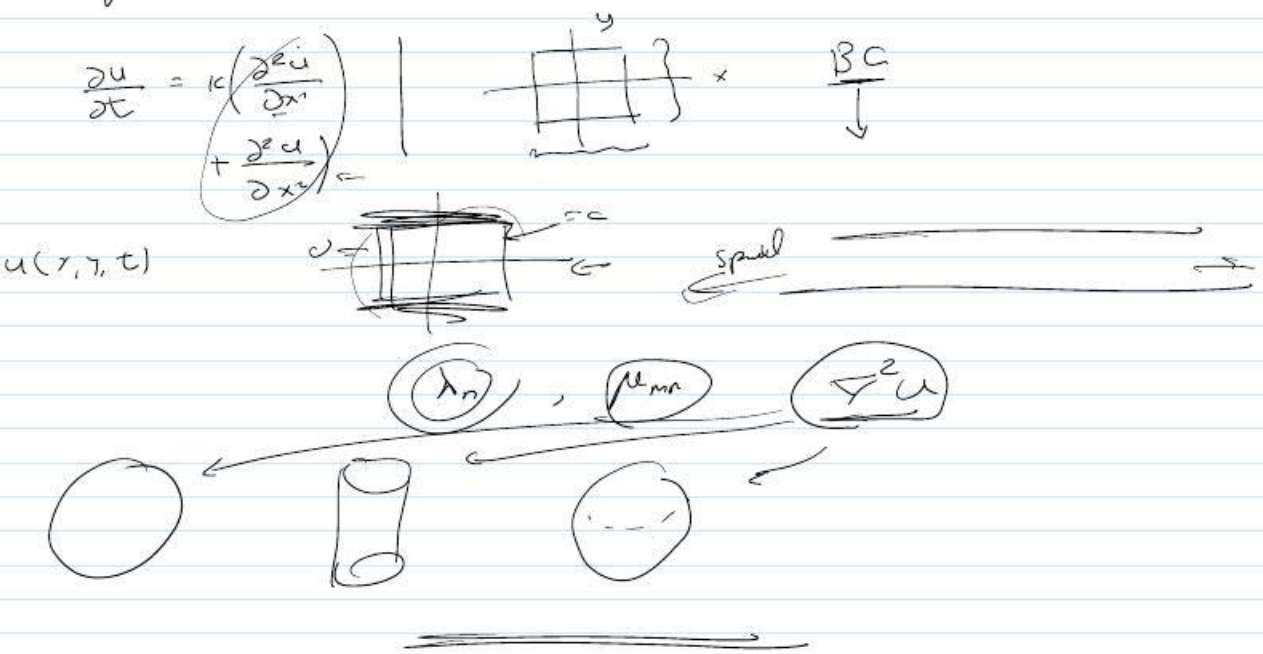
Tutorial

30 December 2023 19:55

PDE in Higher dimensions, unbounded domains

$$\frac{\partial u}{\partial t} = \kappa\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial x^2}\right)$$



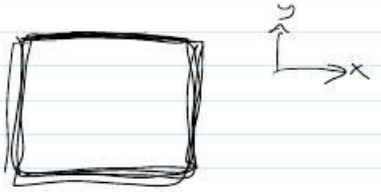
Q.1

$$\frac{\partial^2 u}{\partial t^2} = c^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$$

ODE - 2nd - ②

$$u(0,y,t) = u(L,y,t) = 0$$
$$u(x,0,t) = u(x,H,t) = 0$$

$$u(x,y,0) = \alpha(x,y)$$
$$\frac{\partial u}{\partial t}(x,y,0) = \beta(x,y)$$



$$u(t,x,y) = h(t)\,\phi(x,y)$$

$$\phi\,\frac{\partial^2 h}{\partial t^2} = c^2 h\left(\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2}\right)$$

$$\frac{1}{c^2 h}\frac{\partial^2 h}{\partial t^2} = \frac{1}{\phi(x,y)}\left(\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2}\right) = -\lambda$$

$$\boxed{\frac{1}{c^2 h}\frac{\partial^2 h}{\partial t^2} = -\lambda}$$

$$\boxed{\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = -\lambda\,\phi(x,y)}$$

$$\boxed{\begin{aligned}\phi(0,y) &= \phi(L,y) = 0\\ \phi(x,0) &= \phi(x,H) = 0\end{aligned}}$$

$$\phi(x,y) = \psi(x)\,G(y)$$

$$G(y)\frac{\partial^2\psi}{\partial x^2} + \psi(x)\frac{\partial^2 G}{\partial y^2} = -\lambda\psi G$$

$$\Rightarrow \underbrace{\frac{1}{\psi}\frac{\partial^2\psi}{\partial x^2}}_{X} = \underbrace{-\frac{1}{G}\frac{\partial^2 G}{\partial y^2} - \lambda}_{(y)} = -\mu$$

$$\Rightarrow \boxed{\psi(0) = \psi(L) = 0} \qquad \boxed{\frac{\partial^2\psi}{\partial x^2} = -\psi\mu}$$

$$\psi = c_1 \sin(\sqrt{\mu}\,x) + \cancel{c_2 \cos(\sqrt{\mu}\,x)}^{\,c}$$

$$\psi(0) = c_2 = 0$$

$$\psi(L) = c_1 \sin(\sqrt{\mu}\,L) = 0$$

$$\sqrt{\mu}\,L = n\pi$$

$n = 1, 2, \ldots$

$$\boxed{\mu = \frac{n^2\pi^2}{L^2}}$$

$$-\frac{1}{G}\frac{\partial^2 G}{\partial y^2} - \lambda = \mu_n \quad \Rightarrow \quad \frac{1}{G}\frac{\partial^2 G}{\partial y^2} + \lambda = -\mu$$

$$\Rightarrow \frac{\partial^2 G}{\partial y^2} = -(\mu_n + \lambda)G \qquad \boxed{G(0) = G(H) = 0}$$

($\mu_n + \lambda$: constant)

$$G(y) = \cancel{c_1}^{\,0}\cos\left(\sqrt{\mu_n+\lambda}\,y\right) + c_2 \sin\left(\sqrt{\mu_n+\lambda}\,y\right)$$

$$\sqrt{\mu_n+\lambda}\,H = m\pi$$

$$\mu_n + \lambda = \frac{m^2\pi^2}{H^2}$$

$$\Rightarrow \lambda = \frac{m^2\pi^2}{H^2} - \mu_n = \left(\frac{m^2\pi^2}{H^2} - \frac{n^2\pi^2}{L^2}\right)$$

λ_{mn}

$$\frac{1}{c^2 h}\frac{\partial^2 h}{\partial t^2} = -\lambda_{mn}$$

$$h(t) = c_1\cos(c\sqrt{\lambda_{mn}}\,t) + c_2\sin(c\sqrt{\lambda_{mn}}\,t)$$

$$u = \phi h = G\psi h$$

$$= \left\{c_1\cos(c\sqrt{\lambda_{mn}}\,t) + c_2\sin(c\sqrt{\lambda_{mn}}\,t)\right\}$$

$$= \{ C_1 \cos(c\sqrt{\lambda_{mn}}\, t) + C_2 \sin(c\sqrt{\lambda_{mn}}\, t) \}$$
$$\times \left[K_1 \sin\left(\frac{n\pi}{L}x\right) \right] \left[K_2 \sin\left(\frac{m\pi}{H}y\right) \right]$$

$$u(t,x,y) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \left[A_{mn} \cos\left(c\sqrt{\lambda_{mn}}\, t\right) + B_{mn} \sin\left(c\sqrt{\lambda_{mn}}\, t\right) \right] \sin\left(\frac{n\pi}{L}x\right) \sin\left(\frac{m\pi}{H}y\right)$$

$$\left.\begin{aligned} u(x,y,0) &= \alpha(x,y) \\ \frac{\partial u}{\partial t}(x,y,0) &= \beta(x,y) \end{aligned}\right\}$$

$$u(x,y,0) = \alpha = \sum_{m=1}^{\infty} \left(\sum_{n=1}^{\infty} A_{mn} \sin\left(\frac{n\pi}{L}x\right) \right) \sin\left(\frac{m\pi}{H}y\right)$$

$$K_m$$

$$= \sum_{m=1}^{\infty} K_m \sin\left(\frac{m\pi}{H}y\right) \leftarrow$$

$$K_m = \frac{2}{H} \int_0^H \alpha(x,y) \sin\left(\frac{m\pi}{H}y\right) dy$$

$$= \sum_{n=1}^{\infty} A_{mn} \sin\left(\frac{n\pi}{L}x\right)$$

$$A_{mn} = \frac{2}{L} \int_0^L \left(\frac{2}{H} \int_0^H \alpha(x,y) \sin\left(\frac{m\pi}{H}y\right) dy \right) \sin\left(\frac{n\pi}{L}x\right) dx$$



$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Q2

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = 0$$

$$0 \le r \le 1$$

$$u(1,\theta) = \begin{cases} u_0 & 0 < \theta < \pi \\ 0 & \pi < \theta < 2\pi \end{cases}$$

$$\psi(\theta+2\pi) = \psi(\theta)$$

$\pi < \theta < 2\pi$)

$\psi(\theta+2\pi) = \psi(\theta)$

physical sol.

$u = R(r)\,\psi(\theta)$

$$\psi(\theta)\frac{\partial^2 R}{\partial r^2} + \frac{\psi}{r}\frac{\partial R}{\partial r} + \frac{R}{r^2}\frac{\partial^2 \psi}{\partial \theta^2} = 0 \qquad \times \frac{r^2}{R\psi}$$

$$\Rightarrow \frac{r^2 R''}{R} + \frac{r}{R}R' = -\frac{1}{\psi}\frac{\partial^2 \psi}{\partial \theta^2} = \lambda$$

$$\boxed{\frac{\partial^2 \psi}{\partial \theta^2} + \lambda \psi = 0}$$

$$\psi(\theta) = C_1\cos(\sqrt{\lambda}\,\theta) + C_2\sin(\sqrt{\lambda}\,\theta)$$

$$\psi(\theta+2\pi) = C_1\cos(\sqrt{\lambda}(\theta+2\pi)) + C_2\sin(\sqrt{\lambda}(\theta+2\pi))$$

$$\cos(\sqrt{\lambda}\,\theta) = \cos(\sqrt{\lambda}\,\theta + 2\pi\sqrt{\lambda})$$

$$\boxed{\sqrt{\lambda} = n^2} \qquad \lambda_n = n^2$$

$$\psi(\theta) = \{C_1\cos(n\theta) + C_2\sin(n\theta)\}$$

$$r^2 R'' + rR' - n^2 R = 0 \qquad R = r^m$$

$$m(m-1) + m - n^2 = 0$$

$$m^2 = n^2 \Rightarrow m = \pm n \checkmark$$

$$R(r) = \{C_1 r^{-n} + C_2 r^n\}$$

$\boxed{\lambda > 0}$. $n = 1, 2, \ldots$

$\lambda = 0 \quad \frac{\partial^2 \psi(\theta)}{\partial^2 \theta} = 0 \Rightarrow \psi(\theta) = C_1\theta$

$\psi(\theta+2\pi) = C_1\theta + 2\pi C_1$

$\psi(\theta) = (C_2)$

$\boxed{2\pi C_1 = 0} \Rightarrow C_1 = 0$

$r^2 R'' + rR' = 0$

$\frac{R''}{R'} = -\frac{1}{r} \rightarrow \ln(R') = -\ln(r) + C_1$

$R' = \frac{C_2}{r} \Rightarrow \boxed{R(r) = C_1\ln(r) + C_2}$

$(\lambda < 0)$

$\lambda < 0$

$$u(r,\theta) = \sum_{\lambda} R_\lambda \Psi_\lambda$$

$$= [\cancel{A_0}^{0} \ln(r) + B_0]$$

$$+ \sum_{n=1}^{\infty} \{ \cancel{A_n}^{0} r^{-n} + B_n r^n \}$$

$$\{ C_n \cos(n\theta) + D_n \sin n\theta \}$$

$0 \in D$

$$= A_{r0} + \sum_{n=1}^{\infty} r^n \{ A_n \cos(n\theta) + B_n \sin(n\theta) \} \leftarrow$$

$$u(1,\theta) = \begin{cases} u_0 & 0 < \theta < \pi \\ 0 & \pi < \theta < 2\pi \end{cases}$$

$$u(1,\theta) = A_0 + \sum_{n=1}^{\infty} \{ A_n \cos(n\theta) + B_n \sin(n\theta) \}$$

$$A_0 = \frac{1}{2\pi} \int_0^{2\pi} u(1,\theta)\, d\theta = \frac{u_0}{2\pi} \left[\int_0^{\pi} d\theta \right] = \frac{u_0}{2}$$

$$A_n = \frac{2}{2\pi} \int_0^{2\pi} u(1,\theta) \cos n\theta\, d\theta = \frac{u_0}{\pi} \int_0^{\pi} \cos n\theta\, d\theta$$

$$= \frac{u_0}{\pi} \left[\frac{\sin(n\theta)}{n} \right]_0^{\pi} = 0 \leftarrow A_n = 0$$

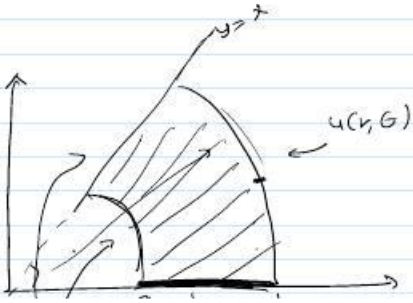
$$B_n = \frac{u_0}{\pi} \int_0^{\pi} \sin(n\theta)\, d\theta = \frac{u_0}{\pi} \left[-\frac{\cos(n\theta)}{n} \right]_0^{\pi} = \frac{u_0}{\pi n} \left[1 - (-1)^n \right]$$

$$\Rightarrow \boxed{u(1,\theta) = \frac{u_0}{2} + \sum_{n=1}^{\infty} \frac{u_0}{\pi n} \left[1 - (-1)^n \right] \sin(n\theta)} \leftarrow$$

Q3

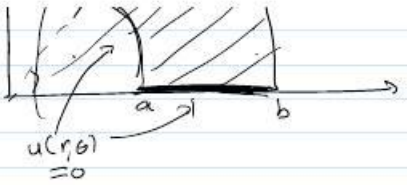
$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0$$

$$u(r,0) = 0 \quad , \quad u(r,\tfrac{\pi}{4}) = 0$$



$u(r,0) = 0 \quad , \quad u(r,\frac{\pi}{4}) = 0$

$u(a,\theta) = 0 \quad , \quad u(b,\theta) = 100$



$$\frac{\partial^2 \Psi(\theta)}{\partial \theta^2} + \lambda \Psi(\theta) = 0 \qquad \Psi(0) = \Psi(\pi/4) = 0 \qquad a < r < b$$

$$\Psi(\theta) = c_1 \sin(\sqrt{\lambda}\,\theta) + c_2 \cos(\sqrt{\lambda}\,\theta) \qquad \lambda > 0$$

$$\sqrt{\lambda}\,\frac{\pi}{4} = n\pi \Rightarrow \sqrt{\lambda} = 4n \Rightarrow \lambda = 16n^2$$

$$R(r) = c_1 r^{-4n} + c_2 r^{4n}$$

$$u(r,\theta) = \sum_{n=1}^{\infty} A_n r^{-4n} \sin(4n\theta) + B_n r^{4n} \sin(4n\theta)$$

$u(a,\theta) = 0, \quad u(b,\theta) = 100$

$$u(a,\theta) = \sum_{n=1}^{\infty} \{A_n a^{-4n} + B_n a^{4n}\} \sin(4n\theta) = 0$$

$$\Rightarrow A_n a^{-4n} + B_n a^{4n} = 0$$

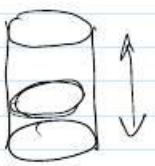
$$A_n = -B_n a^{8n}$$

$$u(r,\theta) = \sum_{n=1}^{\infty} B_n \sin(4n\theta)\{r^{4n} - a^{8n} r^{-4n}\}$$

$$u(b,\theta) = 100 = \sum_{n=1}^{\infty} B_n \sin(4n\theta)\{b^{4n} - a^{8n} b^{-4n}\}$$

$$B_n\{b^{4n} - a^{8n} b^{-4n}\} = \frac{2}{\pi/4}\int_0^{\pi/4} 100 \times \sin(4n\theta)\, d\theta$$

Q4



$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{\partial^2 u}{\partial z^2} = 0$$

$0 \le r \le 1 \qquad 0 \le z \le 1$

$u(1,z) = z$

$u(1,z) = z$ ←

$\left.\frac{\partial u}{\partial z}\right|_{z=0} = \left.\frac{\partial u}{\partial z}\right|_{z=1} = 0$ ←

$u(r,z) = R(r)\,\psi(z)$

$$\frac{1}{R}\frac{\partial^2 R}{\partial r^2} + \frac{1}{rR}\frac{\partial R}{\partial r} = -\frac{1}{\psi}\frac{\partial^2 \psi}{\partial z^2} = \lambda$$

$\lambda > 0$

$$-\frac{1}{\psi}\frac{\partial^2\psi}{\partial z^2} = \lambda \Rightarrow \frac{\partial^2\psi}{\partial z^2} + \lambda\psi = 0$$

$$\Rightarrow \psi = c_1 \cos(\sqrt{\lambda}\, z) + c_2 \sin(\sqrt{\lambda}\, z)$$

$$\left.\frac{\partial \psi}{\partial z}\right|_{z=0} = \left.\frac{\partial \psi}{\partial z}\right|_{z=1} = 0$$

$$\psi' = -c_1\sqrt{\lambda}\sin(\sqrt{\lambda}\, z)$$

$$\psi'(1) = -c_1\sqrt{\lambda}\sin(\sqrt{\lambda}) = 0$$

$\sqrt{\lambda} = n\pi$

$\lambda = n^2\pi^2$ ✓ ∈

$n = 1, 2, 3, \ldots$

$\lambda = 0$

$$\frac{\partial^2\psi}{\partial z^2} = 0 \Rightarrow \psi = c_1 z + c_2$$

$$\psi = c_2$$

$$\psi' = c_1 = 0$$

$$R'' + \frac{1}{r}R' = \underbrace{\lambda}_{n^2\pi^2} R \Rightarrow rR'' + R' - rn^2\pi^2 R = 0$$

Bessel D.E. $\nu = 0$

$$R(r) = c_1 I_0(n\pi r) + c_2 K_0(n\pi r) \qquad 0 \in D$$

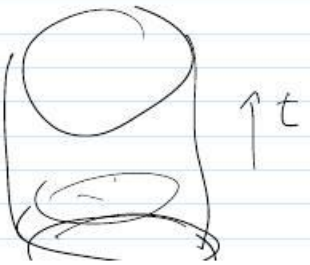
$$u(r,\theta) = \sum_{n=0}^{\infty} A_n I_0(n\pi r)\cos(n\pi z) \leftarrow$$

Q5

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} = \frac{\partial u}{\partial t}$$

$u(2,t) = 100$

$r = 200$ circle



$u(r,0) = \begin{cases} 200 & 0<r<1 \\ 100 & 1<r<2 \end{cases}$



$u(r,t) = v(r,t) + \psi(r)$

$$\left[\frac{\partial^2 v}{\partial r^2} + \frac{1}{r}\frac{\partial v}{\partial r}\right] + \underbrace{\left[\frac{\partial^2 \psi}{\partial r^2} + \frac{1}{r}\frac{\partial \psi}{\partial r}\right]}_{=0} = \frac{\partial v}{\partial t}$$

$v(2,t) + \psi(2) = 100$

$v(r,0) + \psi(r) \begin{cases} 200 & 0<r<1 \\ 100 & 1<r<2 \end{cases}$

$\frac{\partial^2 \psi}{\partial r^2} + \frac{1}{r}\frac{\partial \psi}{\partial r} = 0$

$\psi(2) = 100$

$\psi'' = -\frac{1}{r}\psi'$

$\frac{\psi''}{\psi'} = -\frac{1}{r}$

$\Rightarrow c_1 + c_2 \ln(r)$

$\psi(r) = c_1 = 100$

$\frac{\partial^2 v}{\partial r^2} + \frac{1}{r}\frac{\partial v}{\partial r} = \frac{\partial v}{\partial t}$ $\quad \psi \in [0,\infty)$

$v(2,t) = 0$

$v(r,0) = \begin{cases} 100 & 0<r<1 \\ 0 & 1<r<2 \end{cases}$

$v = h\,R$

$\frac{1}{R}\frac{\partial^2 R}{\partial r^2} + \frac{1}{rR}\frac{\partial R}{\partial r} = \frac{1}{h}\frac{\partial h}{\partial t} = -\lambda$

$\Rightarrow r R'' + R' + \lambda R = 0$ $\quad R(2) = 0$

$\lambda < 0$ Trivial.

$\lambda = 0$ $\quad R = c_1 = 0$ Trivial.

$\lambda > 0$ $\quad R(r) = c_1 J_0(\sqrt{\lambda} r) + c_2 Y_0(\sqrt{\lambda} r)$

$= c_1 J_0(\sqrt{\lambda} r)$

$R(2) = c_1 J_0(2\sqrt{\lambda}) = 0$ $\quad 2\sqrt{\lambda} = \alpha_n$

$J_0(2\alpha_n) = 0$

$R(r) = c_1 J_0(\alpha_n r)$

$\frac{1}{h}\frac{\partial h}{\partial t} = -\alpha_n^2 \Rightarrow h = c e^{-\alpha_n^2 t}$

$R(r) = c_1 J_0(\alpha_n r)$

$$\frac{1}{h}\frac{\partial h}{\partial t} = -\alpha_n^2 \Rightarrow h = c_1 e^{-\alpha_n^2 t}$$

$$v(r,t) = \sum_{n=1}^{\infty} A_n e^{-\alpha_n^2 t} J_0(\alpha_n r) \quad \Big| \quad \underbrace{J_0(2\alpha_n) = 0}_{b=2}$$

$$v(r,0) = \begin{cases} 100 \\ 0 \end{cases} = \sum_{n=1}^{\infty} A_n J_0(\alpha_n r)$$

$$A_n = \frac{2}{2^2 J_1^2(2\alpha_n)} \int_0^2 r\, J_0(\alpha_n r)\, v(r,0)\, dr$$

$$= \frac{100}{2 J_1^2(2\alpha_n)} \left[\int_0^1 r\, J_0(\alpha_n r)\, dr \right] \qquad u = \alpha_n r,\ du = \alpha_n dr$$

$$= \frac{100}{2 J_1^2(2\alpha_n)} \left[\frac{1}{\alpha_n^2} \int_0^{\alpha_n} u\, J_0(u)\, du \right]$$

$$\frac{d}{du}[u J_1(u)]$$

$$= \frac{100}{2 J_1^2(2\alpha_n)} \frac{1}{\alpha_n^2} \left[u J_1(u) \right]_0^{\alpha_n}$$

$$= \frac{100}{2 J_1^2(2\alpha_n)} \times \frac{1}{\alpha_n^2} \alpha_n J_1(\alpha_n)$$

$$= \frac{100}{2 J_1^2(2\alpha_n)} \times \frac{J_1(\alpha_n)}{\alpha_n}$$

$$v(r,t) = \sum_{n=1}^{\infty} \frac{50\, J_1(\alpha_n)}{\alpha_n J_1^2(2\alpha_n)} J_0(\alpha_n r)\, e^{-\alpha_n^2 t}$$

$$u(r,t) = v(r,t) + \underbrace{\Psi(r)}_{100}$$

(a) $\frac{\partial^2 u}{\partial r^2} + \frac{2}{r}\frac{\partial u}{\partial r} = \frac{1}{k}\frac{\partial u}{\partial t}$ + cte ...

Q6

$$\frac{\partial^2 u}{\partial r^2} + \frac{2}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} + \frac{\cot\theta}{r^2}\frac{\partial u}{\partial \theta} = 0$$

$x = \cos\theta$

$$u(r,\theta) = \sum_{n=0}^{\infty} \{A_n r^n + B_n r^{-(n+1)}\} P_n(\cos\theta)$$

@ $r < c$ $\quad 0 < \theta < \pi$ $\quad u(c,\theta)$ $\begin{cases} 50 & 0 < \theta < \pi/2 \\ 0 & \pi/2 < \theta < \pi \end{cases}$

$$u(r,\theta) = \sum_{n=0}^{\infty} A_n r^n P_n(\cos\theta)$$

$n = 2k+1$

$$u(c,\theta) = \sum_{n=0}^{\infty} A_n c^n P_n(\cos\theta)$$

$$f(x) = \sum_{n=0}^{\infty} A_n P_n(x)$$

$$A_n = (n+\tfrac{1}{2}) \int_{-1}^{1} P_n(x) f(x)\, dx$$

$x = \cos\theta \qquad \theta = \cos^{-1}(x)$

$dx = -\sin\theta\, d\theta$

$$= -(n+\tfrac{1}{2}) \int_{\pi}^{0} P_n(\cos\theta) f(\theta) \sin\theta\, d\theta$$

$$= (n+\tfrac{1}{2}) \int_{0}^{\pi} P_n(\cos\theta) f(\theta) \sin\theta\, d\theta$$

$$c^n A_n = (n+\tfrac{1}{2}) \int_0^{\pi} u(c,\theta) P_n(\cos\theta) \sin\theta\, d\theta$$

$$= (n+\tfrac{1}{2}) 50 \int_0^{\pi/2} P_n(\cos\theta) \sin\theta\, d\theta$$

$$c^n A_n = 50 (n+\tfrac{1}{2}) \int_0^{\pi/2} P_n(\cos\theta) \sin\theta\, d\theta$$

$$c^0 A_0 = A_0 = 25 \int_0^{\pi/2} \sin\theta\, d\theta = 25 \left[-\cos\theta\right]_0^{\pi/2} = 25$$

$$c^n A_n = 50 (n+\tfrac{1}{2}) \int_0^1 P_n(x)\, dx = \frac{50(n+\frac{1}{2})}{2n+1} \left[P_{n+1}(x) - P_{n-1}(x) \right]_0^1$$

$1 - 1 = 0$

$$\int P_n(x) = \frac{1}{2n+1} \int \left[P_{n+1}'(x) - P_{n-1}'(x) \right] dx$$

$P_n(1) = 1 \quad \forall\, n > 0$

$$c^n A_n = \frac{50(n+\frac{1}{2})}{2n+1} \left[P_{(n-1)}(0) - P_{(n+1)}(0) \right]$$

$$\begin{cases} P_{2k+1}(0) = 0 \\ P_{2k}(0) = \dfrac{(-1)^k (2k)!}{2^{2k} (k!)^2} \end{cases}$$

$n = 2k+1$

$n-1 = 2k$

$n+1 = 2k+2$

$$c^n A_n = \frac{50(n+\frac{1}{2})}{2n+1} \left[P_{2k}(0) - P_{2(k+1)}(0) \right]$$

$$c^n A_n = \frac{50(n+1/2)}{2n+1}\left[P_{2k}(0) - P_{2(k+1)}(0)\right]$$

$$= \frac{50(2k+1/2)}{2n+1}\left[\frac{(-1)^k (2k)!}{2^{2k}(k!)^2} - \frac{(-1)^{k+1}(2k+2)!}{2^{2k+2}((k+1)!)^2}\right]$$

$$= 25\,\frac{(-1)^k (2k)!}{2^{2k}(k!)^2}\left[1 + \frac{2(k+1)(2k+1)}{2^2 (k+1)(k+1)}\right]$$

$$\left[\frac{2k+2}{2k+2} + \frac{2k+1}{2k+2}\right] = \frac{4k+3}{2k+2}$$

$$c^{2k+1} A_{2k+1} = 25\,\frac{(-1)^k (2k)!(4k+3)}{2^{2k}(2k+2)(k!)^2}$$

$$\mathcal{F}\{f(x)\} = \int_{-\infty}^{\infty} f(x)\, e^{i\alpha x}\, dx = F(\alpha)$$

$$\mathcal{F}^{-1}\{F\alpha\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\alpha)\, e^{-i\alpha x}\, dx = f(x)$$

$$\mathcal{F}_s\{f(x)\} = \int_0^{\infty} f(x)\sin \alpha x\, dx = F(\alpha)$$

$$\mathcal{F}_s^{-1}\{F(\alpha)\} = \frac{2}{\pi}\int_0^{\infty} F(\alpha)\sin \alpha x\, d\alpha = f(x)$$

$$\mathcal{F}\{f^{(n)}(x)\} = (-i\alpha)^n\, \mathcal{F}\{f(x)\}$$
$$= (-i\alpha)^n F(\alpha).$$

$$\mathcal{F}_s\{f''\} = \alpha f(0) - \alpha^2 F(\alpha)$$

$$\mathcal{F}_c\{f''\} = -\alpha^2 F(\alpha) - f'(0)$$

$x \in]-\infty, \infty[$! $x \in [0, \infty[$

Q1 $\mathcal{F}\left(\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}\right)$ $-\infty < x < \infty$ $t > 0$

$u(x,0) = e^{-|x|}$ $\mathcal{F}$ over x

$$\mathcal{F}\left(\frac{\partial u}{\partial t}\right) = \frac{\partial}{\partial t}\mathcal{F}(u(x,t)) = \frac{\partial U(\alpha,t)}{\partial t}$$

$$\mathcal{F}\left(k\frac{\partial^2 u}{\partial x^2}\right) = k(-i\alpha)^2 U(\alpha,t) = -k\alpha^2 U(x,t)$$

$$\frac{\partial U}{\partial t} + k\alpha^2 U = 0 \leftarrow \boxed{U = c_1 e^{-k\alpha^2 t}}$$

$$\mathcal{F}\{U(x,0)\} = \int_{-\infty}^{\infty} e^{i\alpha x} e^{-|x|}\,dx$$

$$= \int_{-\infty}^{0} e^{x+i\alpha x}\,dx + \int_{0}^{\infty} e^{-x+i\alpha y}\,dx$$

$$= \frac{e^{(i\alpha+1)x}}{i\alpha+1}\Big|_{-\infty}^{0} + \frac{e^{(i\alpha-1)v}}{i\alpha-1}\Big|_{0}^{\infty}$$

$\lim_{x\to-\infty} e^{i\alpha x} e^{x} = 0$

$$= \frac{1}{i\alpha+1} - \frac{1}{i\alpha-1} = \frac{2}{1+\alpha^2} = U(\alpha,0)$$

$$U = \frac{2}{1+\alpha^2} e^{-k\alpha^2 t}$$

$$\left\{ U(x,t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-i\alpha x}\frac{2}{1+\alpha^2} e^{-k\alpha^2 t}\,d\alpha \right\}$$

Q8

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \qquad x, y \geq 0$$

$$\left.\frac{\partial u}{\partial x}\right|_{x=0} = 0 \;,\qquad u(x,0) = \begin{cases} 5_0 & 0<x<1 \\ 0 & x>1 \end{cases}$$

$$\mathcal{F}_c(\partial_{x^2} u) + \mathcal{F}_c(\partial_{y^2} u) = 0$$

$$-\alpha^2 U(\alpha,y) - \left.\frac{\partial u}{\partial x}\right|_{x=0}^{0} + \partial_{y^2} U(\alpha,y) = 0$$

$$\frac{\partial U}{\partial y^2} - \alpha^2 U = 0 \Rightarrow U(\alpha,y) = C_1\cosh(\alpha y) + C_2\sinh(\alpha y)$$

$$\frac{\partial U}{\partial y^2} - \alpha^2 U = 0 \Rightarrow U(\alpha, y) = C_1 \cosh(\alpha y) + C_2 \sinh(\alpha y)$$

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$$U(\alpha, y) = C_1 e^{-\alpha y} + C_2 e^{\alpha y}$$

$\lim_{y \to \infty}$

$$U(\alpha, y) = C_1 e^{-\alpha y}$$

$$\mathcal{F}_c\{U(x,0)\} = \int_0^\infty U(x,0) \cos(\alpha x)\, dx$$

$$= 50 \int_0^1 \cos(\alpha x)\, dx = \frac{50}{\alpha} \sin\alpha = U(\alpha, 0)$$

$$U(\alpha, y) = \frac{50}{\alpha} \sin\alpha\, e^{-\alpha y}$$

$$u(x,y) = \frac{100}{\pi} \int_0^\infty \frac{\sin\alpha}{\alpha} e^{-\alpha y} \cos(\alpha x)\, d\alpha$$